

Decision Theory & Black-Litterman

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Winter 2026





Decision Theory

Core idea

- Choose an action
- The future is uncertain
- Good decisions depend on:
 - beliefs about uncertainty
 - objectives/preferences

Example: Should I bring an umbrella?

- Action: bring an umbrella or not
- Uncertainty: will it rain?
- Objective: stay dry without inconvenience

Decision theory gives a framework for making rational choices under uncertainty.



Markowitz vs Black-Litterman

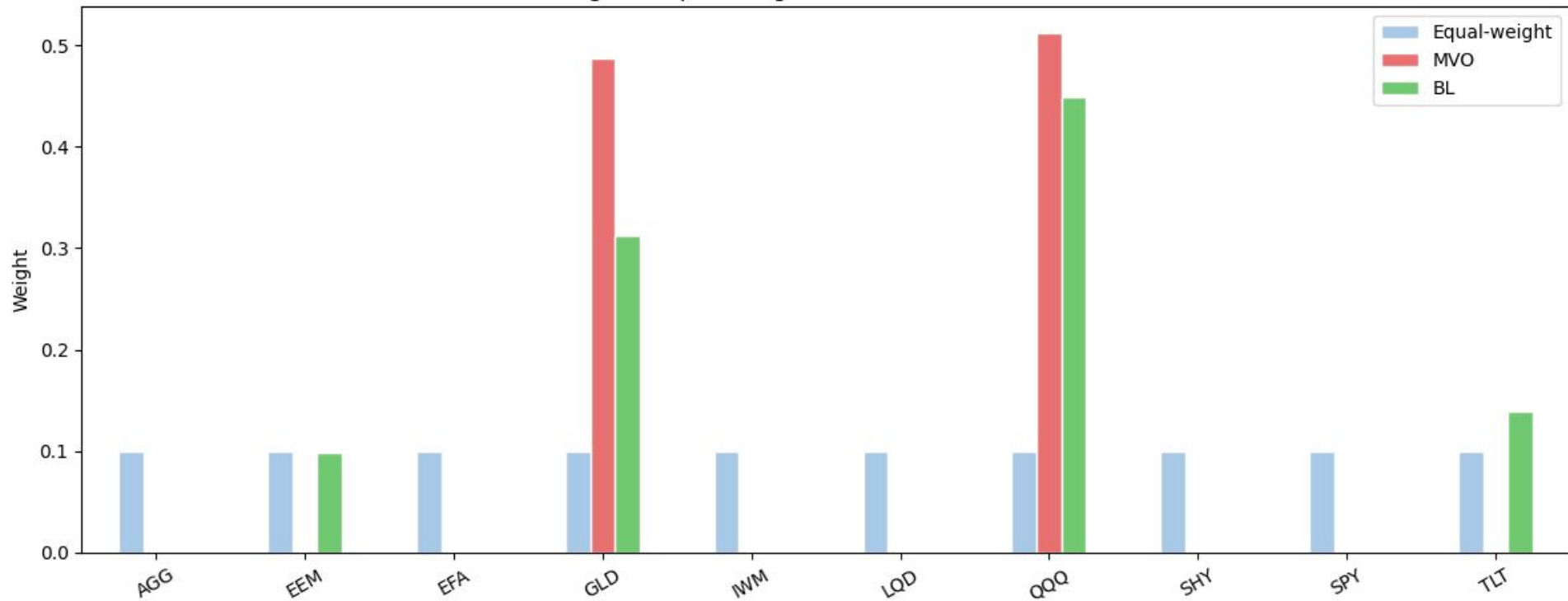
Method	Main Idea	Result
Markowitz (MVO)	Choose portfolio using estimated returns and risk	Very sensitive to noisy return estimates
Black-Litterman (BL)	Start with a prior, then blend in investor views with confidence	More stable and interpretable

Views used in this project:

- QQQ outperforms SPY by 3%
- GLD has expected return 5%

BL can be viewed as a **decision-theoretic approach: prior beliefs + new views + confidence.**

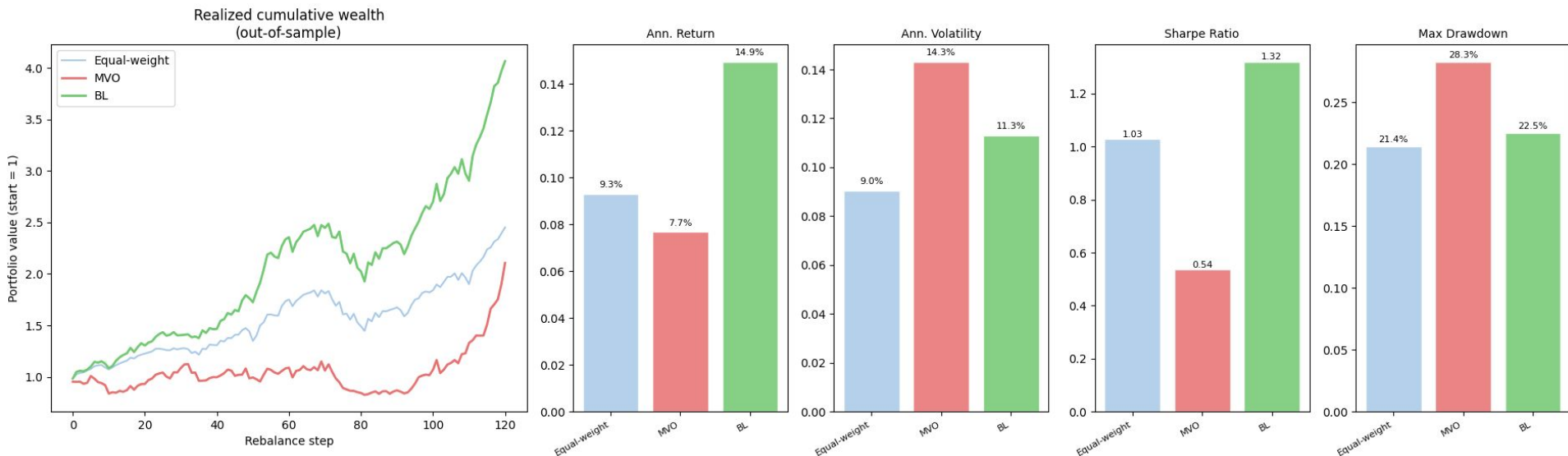
Portfolio weights: Equal-weight vs Markowitz vs Black-Litterman



Notice: MVO concentrates heavily in a few assets, while BL still reflects views but remains more moderated.

- **Equal-weight:** simple diversified baseline
- **MVO:** almost all weight in QQQ and GLD
- **BL:** still tilts toward favored assets, but less extremely

Realized out-of-sample performance comparison



Main Result:

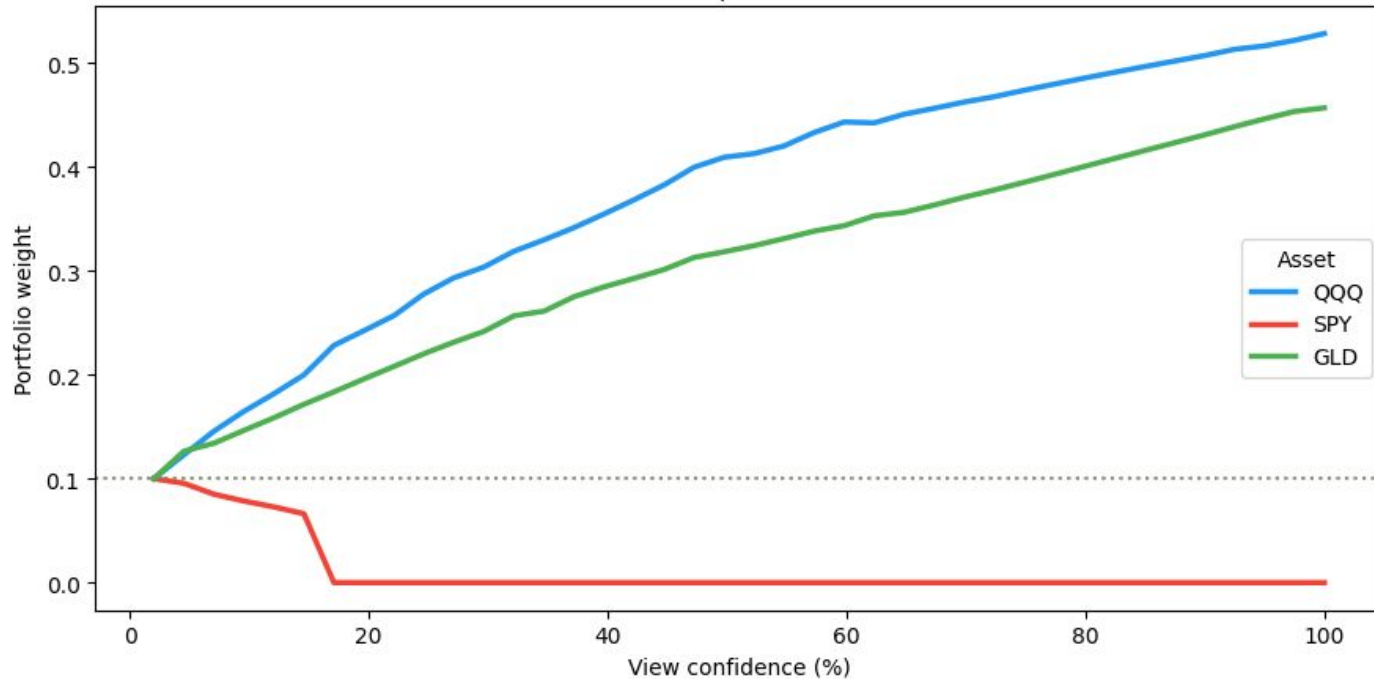
- BL had the highest annualized return
- BL had the highest Sharpe ratio
- BL had lower volatility and drawdown than MVO

annualized return = average return on a yearly basis

Sharpe ratio = return per unit of risk

drawdown = how bad the major losses were from peak to trough

BL confidence sweep: how confidence drives portfolio tilts
(View 1: QQQ-SPY +3% | View 2: GLD absolute 5%)



Higher confidence → stronger tilt toward QQQ and GLD

Lower confidence → portfolio stays closer to the prior

Confidence determines how much views matter